Interface Task Selection & Empirical Problem Framing

Target System Flow

- The system feature that I am going to be targeting is the filtering within the PC Parts Catalog that I have built. This will be evaluating how users transition from the default unfiltered grid to a targeted list using the live keyword search bar and category filter dropdown.

Empirical Rationale

- Bottleneck: In my first base interface that I created the user experiences a lot of friction and frequent "no results" when they were searching for parts which was due to the lack parts in the catalog, strict string matching, and the lack of feedback to the user.
- Redesign: With a new design of the UI I can introduce some type-ahead filtering options for users, similar word matching even if it is not exactly the same, and feedback error handling. This redesign will help reduce task completion time, minimize input errors, and stop dead ends from occurring during selection.

Hypothesis Formulation

Directional Hypotheses (H_1)

- Interface A (Task Efficiency): Users that navigate the newly redesigned UI filtering will complete the item discovery task much faster than users navigating the first base UI.
- Interface B (Effectiveness): Users that navigate the newly redesigned UI filtering will have much less user error when inputting their filtering options when searching for specific hardware items compared to users on the first base UI.

Null Hypotheses (H_0)

- Interface A: There will be no significant difference in task completion time between users using the redesigned interface and users using the first base interface.
- Interface B: There will be no significant difference is the total error frequency between users using the new redesigned interface and users using the first base interface.

Variable Formulation & Operationalization

1. Independent Variables (IVs)

- **Primary Baseline Interface Layout**: The old UI featuring the string matching search, static category tables, and no fallback or loading feedback states.
- **Updated redesigned Interface**: The updated UI featuring the Live Keyword Search Bar, Category Filter Dropdown, Price Sort Toggle, and dynamic empty state error handling.

2. Dependent Variables (DVs)
    - **Efficiency Metric**
        - **Task_Completion_Time_sec**: The time in seconds it takes a user to enter the cursor into the search bar until they find the wanted item and add it to the cart.
    - **Effectiveness Metric**
        - **Error_Count_Total**: The hidden count of the user errors made which includes failed search queries, incorrect category filter selections, or misclicks.
    - **Subjective Satisfaction Metric**
        - **SEQ_Rating**: A scale from 1 to 7 with 1=Very Difficult and 7=Very Easy, which would follow each task to see how users feel.
3. Confounding Variables & Controls
    - **Prior System Familiarity**: Give the user a brief description of both UIs before having them every touch it.
    - **Input hardware**: This will be controlled by having all users use the same mouse and keyboard.
    - **Learning**: This can be controlled by changing the interface design/order for users.
    - **Environmental Distractions**: This will be controlled by testing in a quiet controlled room with standard resolutions and consistent screen brightness.

Experimental Design Architecture & Counterbalancing

Subject Design Selection

- Within-Subjects: By exposing every user to both UIs, we can eliminate the inter-subject demographic variance with a small sample size.

Counterbalancing Scheme

To stop users from learning differently from one another and eliminating fatigue we will use a Balanced Latin Square Counterbalancing Scheme. Users will split evenly across 2 sequences:

1. Sequence 1 (N = 3): Test the old UI first, then make them wait 3 minutes as a reset, then test the new UI.
2. Sequence 2 (N = 3): Test the new UI first, then make them wait 3 minutes as a reset, then test the old UI.

Participant Sampling Plan

- **Target User Personas**: College students and operations professionals with regular experience navigating different component UIs, filtering states, or hardware inventory systems.
- **Inclusion / Exclusion:**
    - *Inclusion*: Familiar with desktop computing workflows and regular web interface navigation
    - *Exclusion*: Prior exposure to similar design to pc part catalogs.
- **Sample Size**: We want to have about 6 users with different backgrounds and experience to fully test our hypothesis and reduce any bias.

Ethical Protocol Safeguards

"We are evaluating the user experience of our new UI for a PC Parts Catalog. You will be testing the efficiency, layout clarity, and overall experience of the catalog while using searches and filtering to find the specific parts. I want to emphasize that we are NOT testing you, we are testing the software. There is no right or wrong answer, and you can pause the session whenever you would like. During the test, you will be asked to find and select specific hardware components using the UI. All data that is collected from this experiment will remain confidential and anonymous. If you have any questions before we begin, please feel free to ask now."